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Product Rating Prediction Using Machine Learning

Project Overview

This project uses Machine Learning to **predict the rating of an e-commerce product** using available product features.

The target variable is **Product_Rating**, and the problem is treated as a **regression problem** because the rating is a numerical value.

Dataset

The dataset contains e-commerce product information such as:

- Product Price
- Discount Rate
- Number of Reviews
- Stock Availability
- Days to Deliver
- Return Rate
- Category ID
- Product Rating

Project Steps

1. Load the dataset
2. Check and clean the data
3. Perform basic EDA
4. Select features and target
5. Convert data into numerical form
6. Split data into training and testing sets
7. Train two regression models
8. Predict product ratings

9. Evaluate the models using:

- MAE (Mean Absolute Error)
- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)

10. Compare the models using a table

11. Identify the model with lower prediction error

Models Used

- **Linear Regression**
- **Random Forest Regression**

Evaluation

The models are compared using MAE, MSE, and RMSE.

For these metrics, **lower error means better prediction performance on the test data.**

Conclusion

The model with the lower error values is selected as the better-performing model for predicting product ratings in this dataset.